



SOC & SIEM • SECURITY MONITORING PROJECT

SIEM Dashboard Setup

SSH Log Monitoring, Failed-Login Tracking & Real-Time Alerting

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What this project covers

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Problem & Objectives

Why SSH brute-force detection matters, and the goals this project set out to prove.

02

Tools & Environment

Kali Linux, VirtualBox, Docker/Splunk – and the hardware limits encountered.

03

Pipeline & Methodology

Ingestion → normalization → dashboard → real-time alerting, step by step.

04

Results & Learnings

What was validated, what went wrong, and what it taught.

Problem Statement & Objectives

THE PROBLEM

Organizations need to detect repeated failed SSH login attempts from a single source in real time – before a brute-force attempt becomes a breach – rather than discovering it after the fact in a post-incident review.

This project also explores how real-world hardware constraints shape the choice of SIEM tooling used for learning and prototyping.

OBJECTIVES

- Design a log ingestion & normalization pipeline for SSH auth data
- Build a dashboard tracking failed logins and unique source IPs
- Rank top offending IPs and targeted usernames
- Configure and validate a real-time brute-force alert
- Evaluate SIEM hardware requirements against the lab machine

Tools & Environment

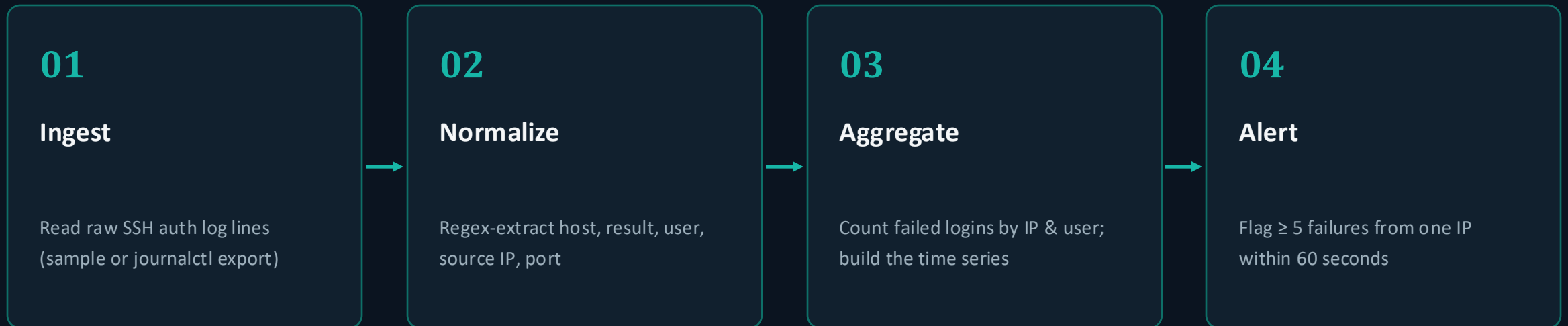
Tool / Software	Platform/Version	Purpose
Kali Linux	2026.2 – VirtualBox VM	Guest OS hosting Docker and generating/exporting SSH authentication logs
Oracle VirtualBox	7.x	Virtualization platform hosting the Kali VM
Docker / Compose	Latest	Container runtime running Splunk Enterprise
Splunk Enterprise	Free trial	SIEM platform: ingestion, dashboard, alerting
Custom Spunk app	Self-built, single file	props.conf field extraction, dashboard XML, saved-search alert rule

LAB MACHINE

Dell inspiron 15 3530

CPU	Intel(R) Core(TM) i7-1355U (1.70 GHz)
RAM	16 GB total
HOST OS	Windows 11
GUEST VM	Kali Linux, 4069 MB base memory

The Detection Pipeline



"Built as a real Splunk Enterprise app (siem_dashboard): field extraction, a Security Monitoring dashboard, and a scheduled saved-search alert — deployed via Docker inside a Kali Linux VM."

Methodology – Key Steps

1

Kali VM Setup

Installed Oracle VirtualBox on the host and imported the Kali Linux 2026.2 VM, configuring it with 4096 MB base memory, 2 processors, and a NAT network adapter. Verified internet connectivity from inside the VM before proceeding.

2

Splunk via Docker

Installed Docker Engine and Docker Compose inside the Kali VM, then deployed Splunk Enterprise as a container using a docker-compose.yml referencing a custom Splunk app folder (siem_dashboard) containing inputs.conf, props.conf, savedsearches.conf, and a dashboard XML.

3

Generating and ingesting SSH logs

Generated sample and live SSH authentication log lines and wrote them to a file monitored by Splunk. Over 2,600 events were successfully ingested into the main index under the siem_ssh sourcetype.

4

Custom Dashboard Built

Built the Security Monitoring dashboard with KPI panels (events ingested, failed logins, unique source IPs), a failed-logins-over-time chart, and top-source-IP / top-targeted-user panels.

5

Ingestion events

2,611 events ingested, 2,231 failed logins, and clear failed-login activity across 5 unique source IPs, with a sharp spike visible in the failed-logins-over-time chart.

6

Alert Validated

Configured a scheduled alert ("SIEM - Brute force from single IP") with the condition "5 or more failed SSH logins from one IP in 5 minutes", a cron-based schedule, and a trigger condition of "Number of Results is > 0". The alert was confirmed Enabled and correctly associated with the siem_dashboard app.

Results & Observations

2611/2231

Ingestion and failed logins

4 / 5 IPs

alert threshold – failures per IP per window

100%

alert fire rate across every simulated attack

KEY FINDING

"The full SIEM pipeline works correctly end-to-end. Four source IPs were identified as brute-force sources, each responsible for 461–611 failed logins — well above the alert's 5-per-5-minutes threshold. The main technical finding was that Splunk's field extractions are app-scoped: an identical query returned 0 results in the default Search app but all 2,231 results inside the siem_dashboard app, a real-world Splunk configuration behaviour that was diagnosed and resolved."

Challenges Faced

VirtualBox import errors when setting up Kali VM

Removed the conflicting VM registration via Machine → Remove, then re-added the VM using Machine → Open with the correct .vbox file, after which the disk showed as Normal and the VM started cleanly

A field-filters search returned 0 events despite Thousands of matching raw text events existing

Diagnosed by first confirming data existed with a raw-text search, then checking the sourcetype and props.conf configuration directly inside the Splunk container, and finally discovering the extraction only applied inside the siem_dashboard app – resolved by running searches from within that app

Splunk web UI became temporarily unresponsive to under combined VM/Docker/browser load

Checked docker ps and restarted the Splunk container with docker restart siem-splunk; service recovered after Splunk's Ansible-based startup completed

Scheduled alert showed "no fired events" even with plenty of qualifying historical data

Learned the alert's time window is relative to real time (last 5 minutes), not the dataset — confirmed the alert's logic, schedule, and enabled status were correct independently via a manual statistics query returning the same qualifying IPs

Learning Outcomes & Conclusion

how to deploy Splunk Enterprise inside Docker, running inside a VirtualBox VM, as a realistic lightweight SIEM lab environment

how Splunk's field extraction (props.conf EXTRACT rules) normalizes raw text logs into structured, searchable fields

Splunk's knowledge objects, including field extractions, are scoped to the app they are defined in – a critical and easily-missed configuration detail when building custom Splunk apps

Learned how to build a SOC-style dashboard aggregating failed-login events into top-offending-IP and top-targeted-user views

how threshold-based, scheduled alerting works in Splunk, including trigger conditions and the distinction between historical data and an alert's real-time scheduling window

systematically diagnose a SIEM pipeline issue by isolating each stage – raw data, sourcetype, field extraction, app context – rather than assuming the first visible symptom was the root cause

CONCLUSION

All objectives were met: a working ingestion & normalization pipeline, a monitoring dashboard with failed-login and top-IP tracking, and a validated real-time alert – adapted to run on constrained hardware without losing the underlying SIEM concepts.

FUTURE SCOPE

Deploy the prepared Splunk config on the 16 GB machine; add web/firewall log sources; add geo-IP lookups; route alerts to email or a webhook.